

 NEXT-GEN CYBER DEFENSE

CyberGuard-AI

AI-Driven Threat Detection & Incident Response System Architecture

Executive Summary

Elevating SOC operational efficiency through hybrid machine learning, continuous threat monitoring, and policy-driven response automation.

The Evolving Threat Landscape

Legacy IDS Limitations

Traditional signature-based security tools rely on known pattern databases, making them highly reactive and ineffective against novel zero-day exploits, evolving malware, and adaptive attack techniques.

Operational Noise & Fatigue

Modern Security Operations Centers (SOCs) are overwhelmed by thousands of daily security alerts. High false-positive rates lead to severe analyst fatigue, delayed response times, and unmitigated risks.

The CyberGuard-AI Solution

- **Continuous AI Monitoring:** Ingests network traffic, system logs, and user payloads in real time.
- **Adaptive Anomaly Detection:** Detects subtle deviations from baseline behavior without relying solely on static rules.
- **Context-Aware Verification:** Evaluates attack severity and confidence metrics before triggering defensive countermeasures.
- **Automated Containment:** Reduces Mean Time to Respond (MTTR) from hours to milliseconds.

Applications of AI in Cyber Security



Key Core Capabilities



Threat Analytics

Monitors network streams and web inputs to identify phishing URLs, XSS payloads, and suspicious payload vectors automatically.



Hybrid ML Models

Combines Isolation Forests for unsupervised anomaly detection with Random Forests for multiclass threat classification.



Automated Response

Executes policy-driven mitigation playbooks, isolating compromised assets and blocking malicious actors in real time.

Model Benchmarks & Metrics

Threat Category	ML Model Applied	Precision	Recall	F1-Score
Phishing & Malicious URLs	Random Forest / XGBoost	96.8%	95.2%	96.0%
XSS & Injection Payloads	Feature-Engineered Classifier	95.4%	94.1%	94.7%
Network Anomaly Detection	Isolation Forest	93.5%	92.8%	93.1%
Combined Mixed Workload	Ensemble Hybrid Stack	95.8%	94.5%	95.1%

Threat Detection Accuracy



CyberGuard-AI significantly outperforms legacy static mechanisms, delivering robust 95.8% overall accuracy on dynamic mixed threat workloads.

Operational Impact Highlights

3.1%

False Positive Rate

drastic Noise Reduction

By lowering the False Positive Rate to just 3.1%, CyberGuard-AI ensures that security teams focus exclusively on high-priority alerts while enabling automated policy enforcement without breaking business-critical operations.

Deployment Roadmap

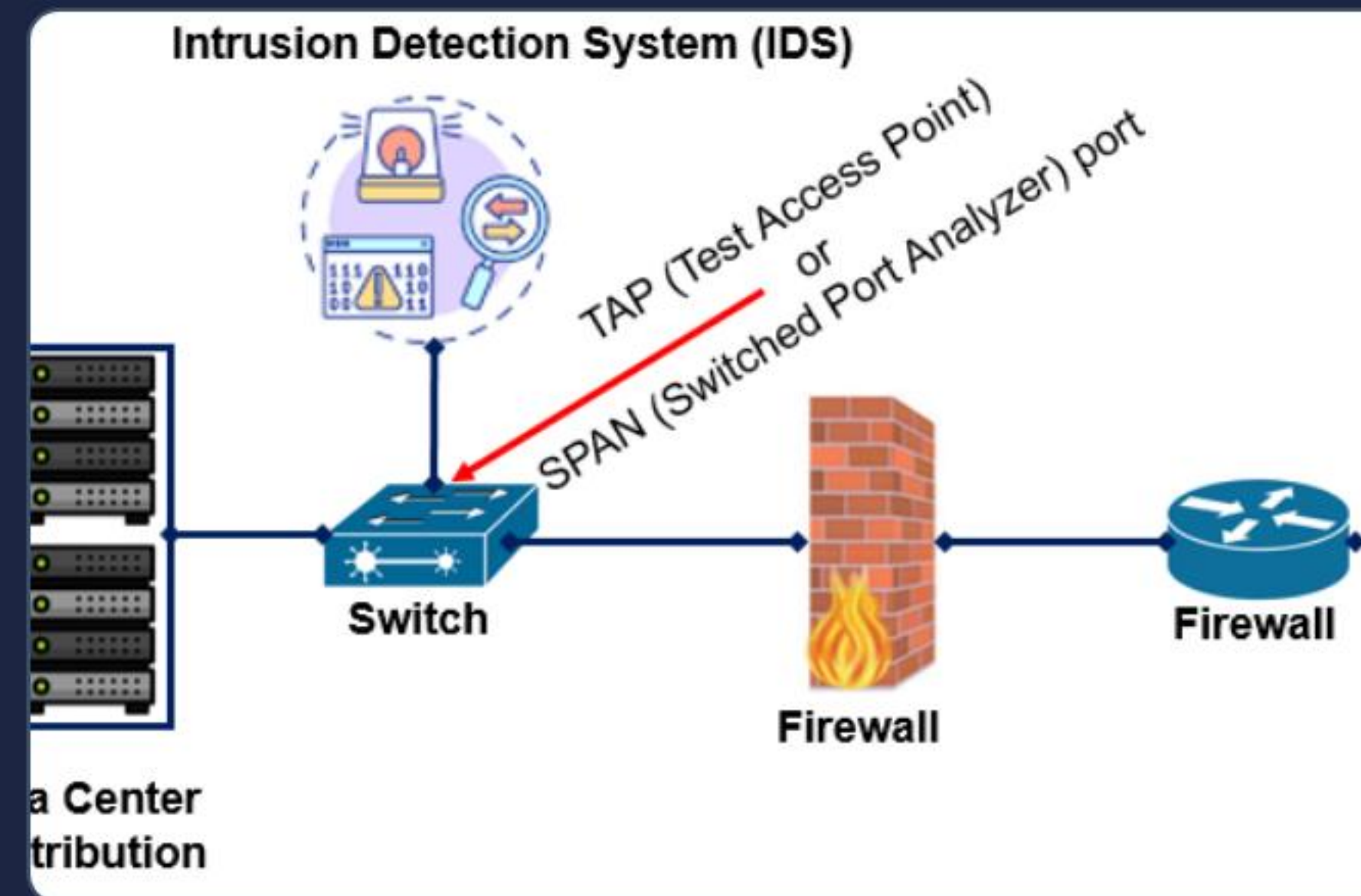


Key Application Scenarios



Phishing Protection

Scans suspicious links and emails to prevent credential harvesting.



Intrusion Defense

Detects unauthorized port scans, brute-force attacks, and lateral movement.



SOC Automation

Provides human-in-the-loop guidance and automated asset isolation.